

Web Programming(CS4032)

Final Exam

Date: June 3, 2025

Total Time: 3 hour

Course Instructor(s)

Total Marks: 40

Mahhek Tahir, Saad Farooq

Total Questions:

Roll No

Section

Student Signature

- Attempt all questions on the answer sheet
- Two A4-sized handwritten cheat sheets, with the Roll Number written at the top of both sheets and stapled together, are allowed.

CLO 4: Develop dynamic, server-side or client-side applications using modern web frameworks and databases

Question 1 -

[20 marks]

You are required to build a basic Library System using Node.js, Express, Mongoose, and JWT.

1. Create Mongoose Models [4 marks]

a. Book Model - models/Book.js [2 marks]

- title: String, **required**, minimum length of **5 characters**
- author: String, maximum length of **30 characters**
- publishYear: Integer, must be between **1900 and 2025** (inclusive)

b. User Model - models/User.js [2 marks]

- username: string, required, minimum length of **10 characters**, unique
- password: string, required
- isAdmin: boolean, default: false

2. Create index.js in the root directory to: [3 marks]

- Connect to a MongoDB database
- Set up Express so that the data received is converted to json.
- Define a basic endpoint to test the API

3. Book Routes – routes/Book.js [6 marks]

- Create the following **Express routes** under /books.
- All routes require authentication using a JWT passed in the Authorization header. Create a middleware that uses verifyToken(token) [already implemented, returns username if token is valid, otherwise returns undefined] to decode the token and get the username. If valid, it allows the controller to run; otherwise, it returns a 401 error in JSON.
- The DELETE route requires an **admin**. To check whether the user is admin or not, create an **adminAuthorization** middleware function. If the user is admin, this middleware will allow the api handler (controller) to run, otherwise it will return an appropriate error message in json with 403 http status code.

National University of Computer and Emerging Sciences

Method	Route	Description
GET	/books	Return a list of all books
POST	/books	Add a new book
GET	/books/search?q=term	Search for books by title or author (case-insensitive, partial match)
DELETE	/books/:id	Delete a book by its ID (admin only)

4. **Implement a controller with the following functions to handle logic for each route: – controllers/Book.js [6 + 1 marks]**

- getAllBooks(req, res)
- addBook(req, res)
- searchBooks(req, res)
- deleteBook(req, res)
- Your controller must handle errors appropriately. [1 mark]

CLO 4: Develop dynamic, server-side or client-side applications using modern web frameworks and databases

Question 2

[10 marks]

You are working on a React frontend that communicates with a backend API hosted at <https://backend.abc.com>. The backend provides the following endpoint:

- GET /is-logged-in:
 1. Returns { "loggedIn": true } if the user is logged in.
 2. Returns { "loggedIn": false } if the user is not logged in.

The frontend includes two components that are already implemented:

- Login (in Login.jsx)
- Home (in Home.jsx)

A partially implemented App.jsx file is also provided. **Your task:**

1. In App.jsx, implement logic to manage the login state using localStorage and React state. Specifically:
 - On page load, check if localStorage contains a loggedIn key.
 - If the key is not present, call the /is-logged-in API, store the result (true or false) in localStorage under the loggedIn key, and update a state variable (isLoggedInState) accordingly.
 - If the key is present, read it from localStorage and use it to update the isLoggedInState without making an API call.
 - The state variable isLoggedInState has been initialized as undefined to indicate that the login status is not yet known.
2. Add the router rendering logic so that:
 - If the user is not logged in, show the Login component at path /login
 - If the user is logged in, show the Home component at path /home.
 - If the login status is not known, i.e., isLoggedInState===undefined, then don't render the /home path at all.
3. In the App component's returned JSX, there is a <div> with its innerText initially set to "Loading...". Create a ref variable to access this <div> and set its innerText (using ref variable) to an empty string ("") once the isLoggedInState is no longer undefined.

National University of Computer and Emerging Sciences

Here is the App.jsx skeleton to complete:

```
import { BrowserRouter, Routes, Route } from "react-router-dom";
import { useEffect, useState } from "react";
import Login from "./Login";
import Home from "./Home";

function App() {

  const [isLoggedInState, setIsLoggedInState] = useState(undefined);
  // On initial load, check if 'loggedIn' exists in localStorage.
  // - If it exists, update isLoggedInState from localStorage (no API call).
  // - If not, call the /is-logged-in API, store the result in localStorage,
  //   and update isLoggedInState accordingly.
  // Write logic here:

  return (
    <>
    { /* update the innerText of the div below as stated in point number 3.*/ }
    <div>Loading...</div>
    <BrowserRouter>
      <Routes>
        { /*
          Only render the /home route if login status is known.
          - If logged in, show the Home component.
          - If not logged in, show the Login component.
          - If status is unknown (undefined), render nothing for /home.
          Write logic here.
        */ }
      </Routes>
    </BrowserRouter>
    </>
  );
}
```

CLO 1: Demonstrate understanding of fundamental web technologies

Question 3

[5 marks]

1. Explain the difference between == and === in JavaScript. **(1)**
2. What is destructuring? How does it simplify code? **(1)**
3. What is the role of res.status(), res.send(), and res.json()? How are they used? **(1.5)**
4. What is the purpose of req.body, req.query, and req.params? Explain with examples. **(1.5)**

CLO 3: Analyze user requirements and web development problems to design solutions

Question 4

[2 + 3 marks]

Part A - If the button in MyComponent below is clicked twice, how many times does "Effect ran" print in total? and why? (1+1 marks)

National University of Computer and Emerging Sciences

```
function MyComponent() {  
  
  const [count, setCount] = useState(0);  
  
  React.useEffect(() => {  
    console.log("Effect ran");  
  }, [count]);  
  
  return (  
    <div>  
      <button onClick={() => setCount(count + 1)}>Click</button>  
    </div>  
  );  
}
```

Part B - Predict the output. (3 marks)

```
function Profile({ user, error }) {  
  return (  
    <div>  
      {error ? (  
        <p>Error: {error}</p>  
      ) : user ? (  
        <>  
          <h2>{user.name || "Guest"}</h2>  
          {user.isActive && user.role === "admin" ? (  
            <p>Admin Access</p>  
          ) : user.isActive ? (  
            <p>Active User</p>  
          ) : (  
            <p>Inactive User</p>  
          )}  
          <p>  
            {user.notifications?.length  
              ? `Notifications: ${user.notifications.length}`  
              : "No notifications"}  
          </p>  
        </>  
      ) : (  
        <p>No user data</p>  
      )}  
    </div>  
  );  
}
```

Assume Profile component above is rendered with different user objects, please predict what will be printed/output for each of the three below:

1. `const user1 = { name: "Anna", isActive: true, role: "admin", notifications: [1,2,3] };`
`<Profile user={user1} />`
2. `const user2 = { name: "Ben", isActive: true, role: "user", notifications: [] };`
`<Profile user={user2} error="Server unavailable" />`
3. `const user3 = { name: "Cara", isActive: false, role: "user" };`
`<Profile user={user3} />`